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|  | Document Type:                                 | Document Code  | RES- PROP-01      |
|                                                                                   | <b>DOCUMENTED INFORMATION</b><br>ISO 9001:2015 | Revision No.   | 00                |
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**(1) Research Program/Project/Study Title:**

**CACAO INDUSTRY IN THE PROVINCE OF CAPIZ: A CASE STUDY**

|                                          |                         |
|------------------------------------------|-------------------------|
| <b>Proponent Name:</b>                   | Ronilo G. Berondo       |
| <b>Academic Rank:</b>                    | Instructor I            |
| <b>Co-Authors (with academic rank):</b>  |                         |
| <b>Campus/Satellite College/Address:</b> | Tapaz Satellite College |
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|                                          | rberondo94@gmail.com    |

Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

|                            |                                                                                                                                                                                                                                            |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Component Title:</b>    | Study 1: Socio-Demographic Profile of Cacao Industry in the Province of Capiz<br><br>Study 2. Issues, Challenges, Difficulties, and Concerns of Cacao Industry.<br><br>Study 3. Need, Support, and Capabilities of Cacao Industry in Capiz |
| <b>Component Leader/s:</b> | N/A                                                                                                                                                                                                                                        |
| <b>Academic Rank:</b>      | N/A                                                                                                                                                                                                                                        |

**(2) Category (please check)**

- ☒ Social Science/Education /Humanities & other related fields  
☐ Natural/Biological Sciences  
☐ Food Research  
☐ Developmental/ Computer/Engineering/ Architecture  
☐ Extension

**(3) Research Center**

[Check the concerned center]

- ☒ SSITRDC  
 Social Science and Industrial Technology Research Development Center  
  
☐ LRDC  
 Livestock Research Development Center  
  
☐ FFRDC  
 Fisheries and Food Research Development Center  
  
☐ CSRDC  
 Crop Science Research Development Center

**(4) Site of implementation**

Municipality/City: Tapaz

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|                                                                                                                                                                                                                                                              |  |                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Province:                                                                                                                                                                                                                                                    |  | Capiz                                                                                                                                                                                                                                                                                                                        |
| Region:                                                                                                                                                                                                                                                      |  | Region VI                                                                                                                                                                                                                                                                                                                    |
| <b>(5) Classification</b><br><br><b>Research:</b> <b>Development:</b><br><br><input type="checkbox"/> Basic <input type="checkbox"/> Pilot Testing<br><input checked="" type="checkbox"/> Applied <input type="checkbox"/> Tech. Promotion/Commercialization |  | <b>(6) Mode of Implementation</b><br><br><input type="checkbox"/> Single Authorship/Proponent<br><input checked="" type="checkbox"/> Collaborative Research (within campus)<br><input checked="" type="checkbox"/> Collaborative Research (intercampus)<br><input type="checkbox"/> Collaborative Research (external agency) |
| <b>(7) Sector/Commodity</b><br><br>Agriculture/Cacao                                                                                                                                                                                                         |  | <b>(8)</b><br><br>Crop and Social Science                                                                                                                                                                                                                                                                                    |

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**(9) Rationale** (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

The Philippine cacao industry is composed mostly of smallholder farmers who supply dry cacao beans to the processors/manufacturers. Manufacturers are mostly small to medium scale enterprises. Some of the larger players are into cocoa powder and cocoa butter. The Philippines both exports and imports cocoa and cocoa products. In 2018, for example, the Philippines produced 7983.20 metric tons of cacao and exported 2732.6 metric tons of cacao beans. Production is highest in the Davao region, representing 82% of total production (Investment Opportunities Presentation, 2020).

In the Province of Capiz, Shared Service Facility (SSF) for cacao processing was launched by the Department of Trade and Industry (DTI), in Barangay Cogon, Panitan, Capiz to address the extensive operations of the local cacao industry. Cacao farmers were provided with machineries for fermentation, drying and sorting of cacao beans. Farmers were also encouraged to plant cacao known as the tropical tree crop that can also a source of wine, vinegar, juice, charcoal. Cacao beans can be use as the main ingredient for beauty, health and wellness. In 2018, the province has already about 34,000 cacao fruit bearing trees out of its estimated 124,000 planted cacao trees. (PIA Capiz, 2022).

However, despite of the many competitive advantages of Cacao in its geographic and climate conditions and proximity to emerging consumer markets, the Philippines' exports remain low. It ranks 72nd in terms of exports and its global market share is less than 0.01%.<sup>3</sup> (The Philippines in the Cocoa-Chocolate Global Value Chain, DTI Policy Briefs, Series No. 2017-09).

In view of the increasing global demand for cacao beans, it's production and commercial production activity, the profiling of Cacao Industry in the province must be established. In order to collect reliable data, provide technological upgrades for Cacao planters/farmers, local processors and micro, small and medium enterprises (MSMEs) in the province and to increase their production capacity, product quality and market competitiveness.

**(9-A) Project Description** (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

**(10) Objectives/Statement of the Problem** (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

The study generally aims to determine the profile of cacao industry in the province of Capiz. Specifically, the study aims to attain the following objectives:

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1. What is the current status of the cacao industry in terms of type of cacao planters/growers (local processors, and micro small and medium enterprises and distributors), type of geographical locations, engagement in product consumption and/or commercialization?
2. What are the issues, challenges, and concerns of cacao farmers/growers?
3. What is the extent of involvement of the cacao growers/farmers in industry planning, implementation, and extension activities of the LGU?
4. What is the extent of the production capacity, product quality, and market competitiveness of the cacao farmer/growers?

**(11) Scientific Basis/Theoretical Framework** (Include concepts, theories and variables that indicate relevant relationships to be tested.

**(12) Review of Literature** (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

Cocoa was first introduced in the Philippines by the Spanish colonizers four centuries ago. The country was the first in Asia to actively produce cocoa beans in the latter part of the 1600s in San Jose, Batangas. By the 1950s, cocoa became a highly profitable commodity as commercial cocoa farms and processing facilities were established. In 1980, cacao production expanded as tax incentives were offered by the Philippine government for large scale investments, particularly in Mindanao where the land and climate were suitable for the crop (Peace and Equity Foundation. A primer on PEF's Priority Commodities: An Industry Study on Cacao. Philippines, 2016. 4. Accessed 26 August 2020).

The average annual cacao consumption is 50,000MT according to a Department of Agriculture (DA). Euro monitor International expects the Philippine chocolate market to grow by 13% by 2017. And that is forecasted to reach 100,000MT by 2020. For the Philippines alone, the local consumption is at 50,000MT annually, but the local supply is only around 10,000MT, making the country a net importer. This looming deficit has given rise to the Philippine Cacao Challenge, which commits the Philippines to produce 100,000MT by the year 2020 and onwards. The focus now is on Mindanao, the southernmost group of islands in the Philippines, which accounts for 90% of the Philippine cacao production, with 80% coming from the Davao Region alone. Majority of Mindanao cacao producers are small farm holdings. Based on aggregate data from Philippine Provincial Agriculturist Offices (PAGROs), Davao Region has more than 20,000 hectares of cacao farms, with Davao City having the largest area of 6,060 hectares (CIDAMI, 2012).

The Philippine Council for Agriculture, Aquatic and Natural Resources Research and Development has identified cacao as one of the priority crops under its Industry Strategic Science and Technology Plans. The plan seeks to further develop the cacao industry through science-based

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solutions and innovations. Using appropriate science and technology interventions, researchers are looking into the need to increase the yield of dried cocoa beans from 500 kilograms per hectare to 800 kilograms per hectare, or an increase equivalent to 60%. The Philippines has already penetrated the Japanese and US markets. In April 2020, the Philippines shipped for the first-time some 20,000 kilograms of Mindanao-sourced premium cacao beans to Belgium, supplied by Mindanao Davao-based premium chocolate brand Auro Chocolate (Export of Philippine Cacao Beans to Europe for the first ever, reported by the Philippine Embassy in Brussels, 2020).

Among the major chocolate manufacturers located in the country are Universal Robina Corporation, Commonwealth Foods, Inc., Goya, Inc., and Columbia International Food Products Inc. Most of these companies are situated in Luzon, with a capacity ranging from 20,000 to 36,000 MT of processed cocoa annually). Universal Robina accounts for 38% of the chocolate market, with Cloud Nine and Nips as its top selling products (7 Department of Agriculture and Department of Trade and Industry, 2017-2022 Philippine Cacao Industry Roadmap).

There are also a number of emerging local artisanal Philippine chocolate brands that make high quality chocolates, including tableas, nibs and cocoa powder, namely Auro, Theo + Philo, Hiraya Chocolates, Risa Chocolates, Tigre Y Oliva, True Chocolate PH, Malagos Chocolate, Coco Dolce, Manila Chocolatier, Kablon Farms, Ginto, Tuesday Baker, and Ralfe Gourmet Chocolate Boutique, among others. Aside from local manufacturers and brands, there are also international companies operating in the country. These companies are engaged in the distribution of imported chocolates and other cocoa-based products as well as in the processing and manufacturing of locally produced beans (Peace and Equity Foundation. A primer on PEF's Priority Commodities: An Industry Study on Cacao. Philippines, 2016. 4. Accessed 26 August 2020).

**(13) Methodology** (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

### **1. Identification and Stratification of Cacao Industry in the Province of Capiz**

A letter from the Office of the University President, Vice President of Research Development and Extension and Director of Research requesting to visit the Department of Agriculture Office of the Province of Capiz will be secured to gather substantial information about the Cacao Industry in the whole province. All data and information to be obtained from the office will be purposely utilized only for research grounds. After successful gathering of relative and relevant information needed from the office, the selected areas for conducting the research will be stratified from the identified municipalities with the greatest number of existing cacao planter/growers/farmers and farms down to its respective barangays.

### **2. Sampling Technique and Procedure**

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The descriptive method of research will be used because; it describes and interprets what is. It concerns with conditions of relationships that exist; practices that prevail; beliefs, processes that are going on; effects that are being felt, or trends that are developing (Calderon and Gonzales, 1993).

All cacao planters/growers/farmers, farms, local processors, micro small and medium enterprises and distributors in the province of Capiz will be the target respondents of the study to obtain reliable and viable research data on profiling of Cacao industry in the territory.

### 3. Data to be Gathered

Study 1: Socio-Demographic Profile of Cacao Industry in the Province of Capiz

Study 2. Issues, Challenges, Difficulties, and Concerns of Cacao Industry.

Study 3. Need, Support, and Capabilities of Cacao Industry in Capiz

### 4. Data Gathering Procedure, Area of Study and Duration

The main instrument in gathering data will be the questionnaires. The instruments will be distributed to the respondents to answer the questions followed by unstructured interview using guide questions. Retrieval of questionnaire will be done immediately to minimize the loss of data. the questionnaire from the respondents the data to be gathered are the personal and work-related characteristics. The study will be conducted at the identified with the most number of existing planters, growers, farmers, local processors, micro small and medium enterprises and distributors in the province estimated to be conducted within three months.

**(14) Major Activities/Work plan** (Enumerate in chronological order the task to be undertaken. Use gantt chart. Below is an example. This can be modified base on your desired Work plan.)

| Research Objectives                                                         | Year 1 |     |     |     |     |     |     |      |      |     | Year 2 |  |  |  |  |  |  |  |  |  |
|-----------------------------------------------------------------------------|--------|-----|-----|-----|-----|-----|-----|------|------|-----|--------|--|--|--|--|--|--|--|--|--|
|                                                                             | Oct    | Nov | Dec | Jan | Feb | Mar | Apr | June | July | Aug |        |  |  |  |  |  |  |  |  |  |
| 1. Preparation and presentation of research proposal                        |        |     |     |     |     |     |     |      |      |     |        |  |  |  |  |  |  |  |  |  |
| 2. Data collection of existing cacao industry in Capiz                      |        |     |     |     |     |     |     |      |      |     |        |  |  |  |  |  |  |  |  |  |
| 3. Preparation of checklist/questionnaires for intended data to be gathered |        |     |     |     |     |     |     |      |      |     |        |  |  |  |  |  |  |  |  |  |

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|----------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|
| 4. Hiring of data entry staff/enumerators                                                                |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 5. Distribution, gathering of data and collection of checklist/survey questionnaires on identified areas |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 6. Recording, analysis and interpretation of the data gathered                                           |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                          |  |  |  |  |  |  |  |  |  |  |  |  |  |

**(15) Expected Output** (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

- Cacao Processed Products

**(15) Technology Transfer Activity Design** (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

**(16) Targets Beneficiaries** (Who the clienteles are and what are the expected outcome/effects of the use of the projects output)

- Future researchers
- Extension providers

**(17) Personnel Requirement**

| Name                     | Role<br>(Program, Project, Study Leader) | Field of Specialization |
|--------------------------|------------------------------------------|-------------------------|
| Ronilo G. Berondo        | Proponent                                | Social Science          |
| Rey C. Castillon         | Co-Author                                | Social Science          |
| Rolly G. Gimoto          | Co-Author                                | Social Science          |
| Louis Placido F. Lachica | Co-Author                                | Social Science          |
| Leo Andrew B. Bicular    | Co-Author                                | English/Literature      |

**Percent time devoted to the Program/ Project/ Study**

**Program/Project Leader (in hours):**

**Program/Project Members (in hours):**

**(18) Budgetary Requirement**

|                                                                                   |                 |                                                |                |                          |
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| SOURCES OF EXPENSES                                                                       | QUANTITY | UNIT    | PRICE PER UNIT | TOTAL            |
|-------------------------------------------------------------------------------------------|----------|---------|----------------|------------------|
| <b>SUPPLIES</b>                                                                           |          |         |                |                  |
| Printer (Epson L3150)                                                                     | 1        | pc      | 9,000          | 9,000            |
| Digital camera (Canon EOS 600D)                                                           | 1        | pc      | 18,000         | 18,000           |
| Digital Weighing scale                                                                    | 1        | pc      | 2,000          | 2,000            |
| A4 size bond paper                                                                        | 10       | reams   | 350            | 3,500            |
| Black and colored inks                                                                    | 4        | bottles | 300            | 1,200            |
| Tape measure                                                                              | 5        | pcs     | 50             | 250              |
| Recording Notebook                                                                        | 1        | pc      | 80             | 80               |
| Ballpen                                                                                   | 10       | pc      | 30             | 300              |
| SUBTOTAL                                                                                  |          |         |                | <b>34,250.00</b> |
| <b>Personnel Services</b> (For studies needed the Research Assistant, informant/s, guide) |          |         |                |                  |
| Research Assistant                                                                        | Days     | 90      | 500            | 45,000.00        |
| Travel Allowance                                                                          | Days     | 90      | 150            | 13,500           |
| SUBTOTAL                                                                                  |          |         |                | <b>58,500.00</b> |
| <b>TOTAL EXPENSES</b>                                                                     |          |         |                | <b>92,750.00</b> |

**(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)**

Peace and Equity Foundation. A primer on PEF's Priority Commodities: An Industry Study on Cacao. Philippines, 2016. 4. Accessed 26 August 2020.

Investment Opportunities Presentation.

The Philippines in the Cocoa-Chocolate Global Value Chain", DTI Policy Briefs, Series No. 2017-09.

Peace and Equity Foundation. A primer on PEF's Priority Commodities: An Industry Study on Cacao. Philippines, 2016. 4. Accessed 26 August 2020.

Cacao/Tablea, Securing The Future of Philippine Industries, [www.industry.gov.ph](http://www.industry.gov.ph).

Export of Philippine Cacao Beans to Europe for the first ever", reported by the Philippine Embassy in Brussels. 7 Department of Agriculture and Department of Trade and Industry, 2017-2022 Philippine Cacao Industry Roadmap.

8Peace and Equity Foundation. A primer on PEF's Priority Commodities: An Industry Study on Cacao. Philippines, 2016. 4. Accessed 26 August 2020.

**(20) Capsule Curriculum Vitae (One-page brief C.V. only)**

Name : Ronilo G. Berondo  
Date of Birth : March 9, 1985  
Citizenship : Filipino  
Gender : Male  
Civil Status : Married

|                                                                                   |                                                |                |                   |
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|-----------------|---|--------------------------|
| Academic Rank   | : | Instructor I             |
| Present Station | : | Tapaz Satellite College  |
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